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LIGHTWEIGHT IMAGE SUPER-RESOLUTION PREPROCESSOR FOR JPEG COMPRESSION

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ABSTRACT

This paper proposes a lightweight image super-resolution (SR) pre-processing method for JPEG compression. Our method reduces computational costs for super-resolution and ensures consistent image quality by avoiding unknown super-resolution algorithms on the user side, which may cause inconsistencies in the quality of the delivered images. Specifically, the proposed method contains an SR model with only 2.8K parameters, followed by the JPEG encoder. For the lightweight SR model, a Separable Structural Reparameterization (SSR), which consists of a Cross-Scale Directional Reparam (CDR) module that improves spatial learning and a Channel Reparam Integration (CRI) module that enhances feature interaction, is proposed to improve the capability of the model for information enhancement. Furthermore, we introduce a JPEG compression-aware training strategy to make the super-resolution process aware of the subsequent JPEG compression. This strategy uses a differentiable JPEG codec for gradient backpropagation during training and switches to the standard JPEG codec for deployment. Extensive experiments have been conducted to validate the effectiveness of our method.

***Index Terms*—** JPEG compression, pre-processing, single image super-resolution

1. INTRODUCTION

In contrast to performing super-resolution on low-resolution images as a post-processing method, providing super-resolution before delivering images to the end user is gaining increasing attention. This approach either reduces computational costs on the user side or ensures consistent quality of the delivered images. At the same time, the preprocessed images are often compressed using image compression methods, such as the widely adopted JPEG (Joint Photographic Experts Group) [1]. However, applying super-resolution to low-resolution images followed by lossy compression presents a significant challenge in maintaining the visual quality of these images, particularly in resource-constrained applications.

In practical applications, while traditional interpolation algorithms may demand fewer computational resources and can be straightforward to implement, deep learning methods present clear advantages, such as their ability to learn and preserve high-frequency details adaptively. A key strength of deep learning-based super-resolution algorithms in our scenario lies in their potential to be jointly trained to perceive JPEG compression. This capability enables them to mitigate the information loss that is commonly associated with lossy compression, making them effective in improving the visual quality of images.

Despite several previous lightweight deep learning-based SISR models [2, 3] have gained excellent performance, these approaches are limited for our target scenario, where a lightweight super-resolution architecture with the compression-aware ability to effectively perceive subsequent compression tasks is required to meet the stringent requirements of resource-constrained devices.

To tackle these challenges, we propose a lightweight JPEG compression-aware super-resolution network. We propose the Separable Structural Reparameterization (SSR) to enhance the model's information enhancement capabilities. The SSR comprises two key components: the Cross-Scale Directional Reparam (CDR) module, which improves the utilization of spatial information, and the Channel Reparam Integration (CRI) module, which enhances feature interaction across different channels. This design enables more effective extraction and representation of high-frequency details, resulting in improved image quality and clearer predictions. Additionally, to meet practical application requirements, we introduce a dynamic upsampling kernel [4] to replace the traditional pixel shuffle module, as the latter often leads to parameter redundancy and fails to align with our stringent requirements for a lightweight network.

Moreover, by employing structural reparameterization [5] and weight-sharing techniques [6], we develop a lightweight super-resolution network that maintains an impressively low parameter count of only 2.8 K. To ensure that the preprocessing super-resolution network effectively accounts for the subsequent JPEG compression during training, we in-

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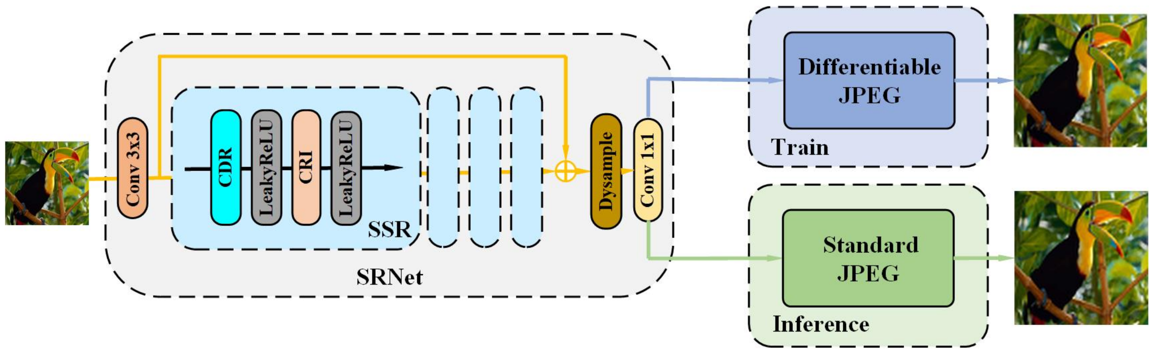

Fig. 1. The entire framework is trained end-to-end by back-propagating error signals through the differentiable JPEG module.

troduce a JPEG compression-aware training strategy. This strategy utilizes a differentiable JPEG codec [7] for gradient backpropagation while switching to the standard JPEG codec for inference during deployment. The effectiveness of our method is validated through extensive experiments.

The remainder of this paper is organized as follows: Section 2 presents the proposed method in detail. Section 3 evaluates the performance of the proposed model through comprehensive experimental results, performance comparisons with baseline methods, and an ablation study. Finally, the paper concludes in Section 4.

2. PROPOSED METHOD

2.1. Overview of Architecture

The overall architecture of our proposed method is illustrated in Fig. 1. Our approach employs a super-resolution network (SRNet) followed by a differentiable JPEG module [7] as our primary training framework. Given a low-resolution (LR) input image $x \in R^{3\times H\times W}$, our SRNet first generates a super-resolution (SR) representation. This representation is then converted into a compressed image $y \in R^{3\times sH\times sW}$, where s is the upscaling factor, using a differentiable JPEG module, which supports both forward and backward gradient propagation. This design enables end-to-end optimization, ensuring the SRNet accounts for compression effects during training.

In the testing phase, the jointly trained model is evaluated by coupling it with a standard JPEG codec to measure the rate-distortion performance.

2.2. SR Network Architecture

The structure of the super-resolution (SR) network is illustrated in Fig. 1. The network comprises two main components: Separable Structural Reparameterization (SSR) modules and an upsampling layer. Initially, a convolutional layer with a kernel size of 3×3 is applied to produce the shallow feature X_0. Subsequently, multiple stacked SSR modules are utilized to extract cross-scale deep features X_1. Each SSR module consists of a Cross-Scale Directional Reparam (CDR) layer and a Channel Reparam Integration (CRI) layer. To reconstruct the high resolution (HR) image, we employ a global residual connection [8], which facilitates efficient gradient flow and simplifies the training process by directly propagating the information from the input image to the output. For rapid and effective upsampling, we introduce an upsampling layer composed of only two components: a dynamic upsampler [4] and a 1×1 convolutional layer. This compact yet powerful design ensures that the final output is accurately resized to the target resolution while maintaining computational efficiency. The network, as a whole, can be expressed in the following formulation:

$$X_0 = F_0(x), \tag{1}$$

$$X_1 = F^{n}_{\text{SSR}}\big(F^{n-1}_{\text{SSR}}(\dots F^{0}_{\text{SSR}}(X_0)\dots)\big), \tag{2}$$

$$z = F_{\text{up}}(X_0 + X_1), \tag{3}$$

where x is the input image, z is the output SR image. $F_0(\cdot)$ denotes the first convolution operation, $F^{n}_{SSR}(\cdot)$ denotes the n-th SSR module, and $F_{up}(\cdot)$ is the upsample module. To increase the capacity of the model without introducing additional parameters, we employ a weight-sharing mechanism [6] to improve the depth of the model. The weights are initially established in the first reparameterization block and subsequently propagated to the following blocks.

2.2.1. Cross-Scale Directional Reparam

Compared to traditional approaches that rely on large-kernel convolutions and pooling operations to expand the receptive field, we introduce a lightweight and efficient alternative designed to reduce model complexity while maintaining performance. Specifically, we replace standard convolutions with depthwise convolutions, significantly reducing the number of parameters and computational costs. Our method employs a multi-branch structure with reparameterization [5], enabling the network to merge these branches into an equivalent single convolution during inference, thereby enhancing efficiency and flexibility.

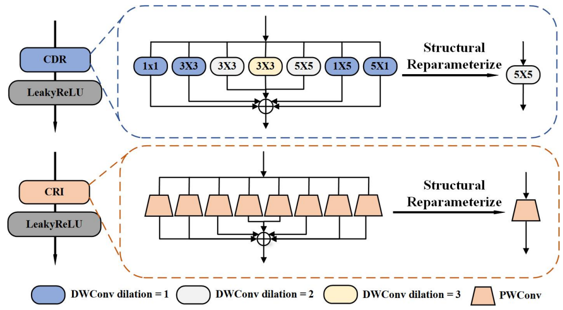

Fig. 2. Illustrations of the Separable Structural Reparameterization (SSR).

As illustrated in Fig. 2, we improve cross-scale information capture by using dilated convolutions. By combining larger sparse kernels with small-scale convolutions in parallel, the network achieves a broader receptive field while maintaining a strong focus on central information [9]. Furthermore, we leverage asymmetric convolutions to extract multi-directional features, allowing the model to better handle variations such as scaling and flipping. This combination of techniques improves the adaptability of the network, enabling more effective feature extraction for high-quality image reconstruction.

2.2.2. *Channel Reparam Integration*

The Cross-Scale Directional Reparam (CDR) module is specifically designed to expand the receptive field and explore a broader range of pixel information by utilizing depthwise convolutions. However, depthwise convolutions inherently lack inter-channel feature connections, which significantly limits their ability to fully integrate and leverage information distributed across different channels. To overcome this limitation, we introduce the Channel Reparam Integration (CRI) module, which is specifically tailored to enhance inter-channel interactions and enrich feature representation.

CRI achieves this enhancement through a series of point-wise convolutions, which are highly efficient in aggregating channel-wise information. These point-wise convolutions enable the network to model complex interdependencies between channels, effectively compensating for the inherent limitations of depthwise convolutions. Additionally, CRI leverages the additive property of convolution operations to consolidate multiple parallel layers with identical kernel sizes into a single equivalent layer during inference. This reparameterization not only preserves computational efficiency but also significantly boosts the network's capacity to capture diverse and comprehensive feature representations. By effectively complementing the broader receptive field and directional perception capabilities of the CDR module, CRI ensures a well-balanced architecture that excels in both efficiency and feature extraction capability.

2.3. Training Strategy for JPEG-aware SR

By incorporating a differentiable JPEG [7] module into the training pipeline, our approach enables the super-resolution network to effectively account for compression artifacts, optimizing the balance between resolution enhancement and post-compression quality. The differentiable JPEG module simulates the compression process while preserving gradient flow, allowing the network to adapt its outputs to real-world compression scenarios.

In our method, the super-resolved (SR) images produced by the network are passed through the differentiable JPEG module, generating compressed images that are compared against high-resolution (HR) images to compute distortion. This ensures that the network is trained to minimize both perceptual loss and compression-related degradation. To achieve optimal performance, we adopt a combination of loss functions. The Charbonnier pixel loss [10] provides robust pixel-level supervision, effectively addressing outliers and ensuring smooth training. Additionally, a Fourier transform-based loss function is introduced to focus on high-frequency components, which are critical for preserving fine details and textures often compromised during compression. This dual-loss strategy enables the network to excel in both spatial and frequency domains, leading to an overall improvement in image quality. The final JPEG optimization loss function is formulated as follows:

$$\mathcal{L} = \mathcal{L}_{Charb}(D_{JPEG}(x), \hat{x}) + \lambda \cdot \|\mathcal{F}(D_{JPEG}(x)) - \mathcal{F}(\hat{x})\|, \tag{4}$$

where $\mathcal{L}_{Charb}(\cdot)$ is the Charbonnier pixel loss, $\mathcal{F}(\cdot)$ is the Fourier transform function. $\| \cdot \|$ denotes the L1-norm. x denotes reconstructed SR image, $\hat{x}$ denotes the HR image. $D_{JPEG}(x)$ is the compression operation using a differentiable JPEG [7].

3. EXPERIMENTAL RESULTS

3.1. Implementation Details

We train our model on the DIV2K [11] dataset, which consists of 800 high-resolution images for training and 100 images for validation. For evaluating rate-distortion performance, we test our model on seven standard benchmark datasets: Set5 [12], Set14 [13], BSD100 [14], Urban100 [15], Manga109 [16], CLIC [17], and Kodak [18]. To simulate the low-resolution images obtained by resource-constrained devices, low-resolution (LR) images are generated by downsampling high-resolution (HR) images using bicubic interpolation. We randomly crop HR patches of size 256×256 from the ground truth images and set the mini-batch size to 64. The Adam optimizer [19] is used with parameters β_1 = 0.9 and β_2 = 0.999. The initial learning rate is set to 1×10^{-3} and decays by half every 100 epochs to gradually refine the model. The model converges after training for nearly 1000 epochs.

Table 1. BD-rate comparison (lower is better) of PSNR/SSIM against Bicubic on various datasets.

Method	Kodak	CLIC	Set5	Set14	BSD100	Urban100	Manga109
Lanczos4	-1.97% / -1.11%	-3.48% / -2.93%	-1.87% / -1.82%	-3.34% / -3.11%	-3.87% / -3.13%	-4.99% / -4.89%	-4.16% / -3.78%
SR-LUT	-5.53% / -7.34%	-8.59% / -11.27%	-7.72% / -8.17%	-10.79% / -11.98%	-10.10% / -12.11%	-13.35% / -14.99%	-14.60% / -14.05%
HiT-SR	-10.21% / -12.49%	-18.44% / -20.38%	-14.69% / -15.15%	-22.58% / -22.88%	-20.76% / -19.57%	-22.33% / -23.56%	-31.19% / -30.02%
Ours	-6.68% / -7.66%	-10.15% / -11.93%	-8.28% / -8.87%	-11.91% / -12.25%	-12.85% / -12.45%	-13.47% / -15.22%	-17.79% / -16.46%

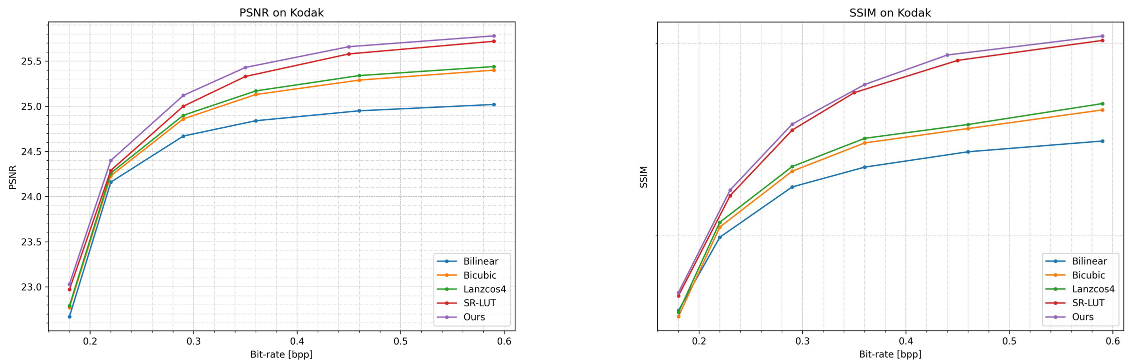

Fig. 3. The rate-distortion performance of Bilinear, Bicubic, Lanczos4, SR-LUT and our method on the Kodak dataset.

Table 2. Model efficiency comparison. Runtime includes preprocessing and JPEG compression.

Method	Params (K)	Runtime (ms)
Bicubic	-	0.9
Lanczos4	-	1.7
SR-LUT	18.0	213*
HiT-SR	792.0	731
Ours	**2.8**	**33**

* The runtime of SR-LUT is evaluated using the official NumPy implementation.

3.2. Benchmark Evaluations

Given the significantly smaller parameter size of our network, it is difficult to make comparisons with other state-of-the-art lightweight SR. We compared traditional interpolation methods like Bicubic and Lanczos. In addition, we also included SR-LUT [2] and HiT-SR [20] as our comparison methods, as they are among the most lightweight learning based methods as far as we can obtain.

We quantitatively evaluate rate-distortion performance using BD-rate [21] with bicubic as the anchor. The bitrate is calculated by dividing the actual file size of the JPEG-compressed image by the image area, ensuring a fair comparison across methods. Inference time and model parameters are measured by upscaling 64×64 images to 256×256 on an Nvidia RTX 3090 GPU. As shown in Table 2, our method outperforms most comparison methods in rate-distortion performance while maintaining a lightweight structure, with only 2.8K parameters and a runtime of 33ms. While our method adds more computational complexity than traditional interpolation methods like Bicubic and Lanczos4, the improvement, though modest, should not be seen as a drawback. As shown in Table 1 with HiT-SR, although our method does not surpass HiT-SR in absolute performance, it uses only 1/280 of the parameters (792K vs. 2.8K) and achieves over 20× faster inference (731ms vs. 33ms), highlighting its advantage in lightweight design and real-time applications. Our method shows consistent performance across datasets and metrics, indicating strong generalization.

Our observations indicate that when the bit per pixel (bpp) exceeds 0.3, a significant deceleration is observed on PSNR, as depicted in Fig. 3. Unlike traditional compression, where PSNR is determined by comparison with the original image, the characteristics of the super-resolution model make it impossible for the reconstructed image to be completely identical to the original. Consequently, visual evaluation metrics do not exhibit pronounced sensitivity to variations in bpp.

Fig. 4 presents visual results of the pre-processed 4× upscaling super-resolution network after JPEG compression. Our method preserves more texture details and achieves sharper edges, leading to higher model fidelity. For example, the wave layers are more distinct, and the tread patterns on the tire edges are closer to the original image.

3.3. Ablation Studies

3.3.1. Effectiveness of Differentiable JPEG Module

To validate the effectiveness of the differentiable JPEG module during training, we conduct an ablation study by removing this module. We evaluate the performance on Kodak [18],

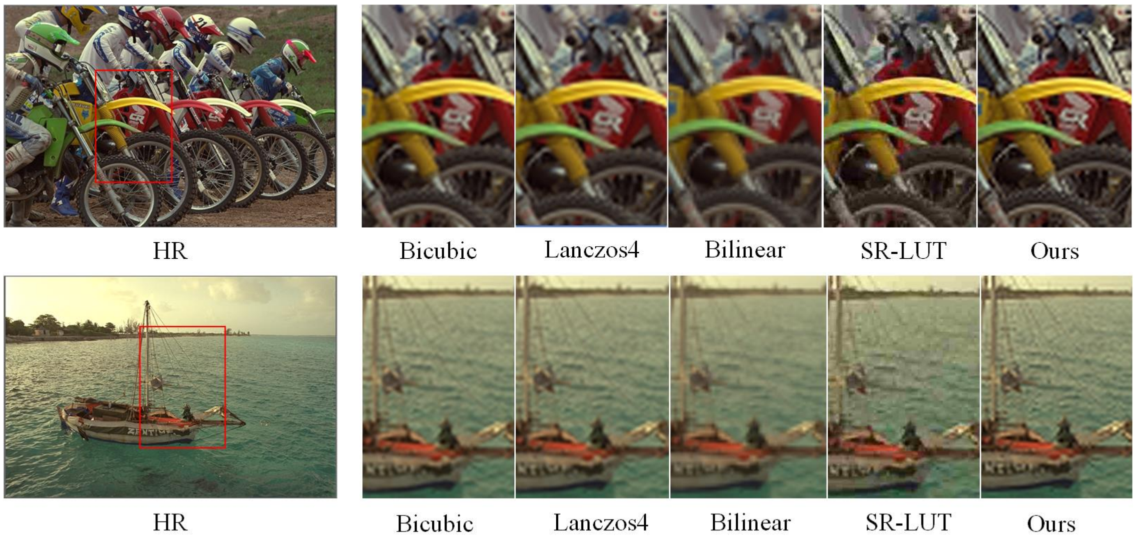

Fig. 4. Visual comparisons of different methods. Parts of the areas are zoomed in with red boxes.

Table 3. BD-rate comparison measured by PSNR of w/wo differentiable JPEG joint training, different modules without structural reparameterization: only without CDR (w/o CDR reparam) and only without CRI (w/o CRI reparam).

Method	Kodak	Set5	Set14	Param (K)
w/ joint	-6.68%	-8.28%	-11.91%	2.8
w/o joint	-5.43%	-6.41%	-10.16%	2.8
w/o CDR reparam	-3.51%	-4.62%	-8.12%	2.8
w/o CRI reparam	-3.76%	-4.88%	-8.28%	2.8

Set5 [12], and Set14 [13]. As shown in Table 3, results with the compression-aware module (w/ joint) outperform those without it (w/o joint) across all evaluation metrics. By incorporating the compression operation during training, the model learns to handle compressed images through forward propagation, computes loss using these images, and optimizes via backpropagation. This approach effectively leverages noise information to enhance model performance.

3.3.2. *Effectiveness of SSR*

To verify the rationality of using structural reparameterization for separate SSR module training, we perform an ablation study by applying structural reparameterization to one part while not using it for the other, as shown in Table 3. We further assess the impact of cross-scale and multi-directional multi-branch perception by substituting the SSR module with a parameter-matched depthwise separable convolution. With bicubic interpolation as the reference, the BD-rate (PSNR) gains under JPEG compression reach -2.44% (Kodak), -3.70% (Set5), and -7.53% (Set14). This confirms that multi-branch reparameterization with structural diversity provides substantial benefits for lightweight architectures.

3.3.3. *Effectiveness of Dysample Module*

In our exploration of lightweight super-resolution methods, we find that the pixel shuffle method for upsampling does not meet our practicality requirements. To verify its impact, we implement an alternative version using pixel shuffle, which achieves BD-rate (PSNR) gains of -6.71% (Kodak), -8.29% (Set5), and -11.93% (Set14), using bicubic interpolation as the anchor under JPEG compression. Although the performance is comparable to that shown in Table 3, the pixel shuffle variant requires 16.4K parameters—substantially more than our configuration. Our experimental analysis leads us to adopt the dysample-based upsampling kernel, as it delivers competitive accuracy while maintaining minimal parameters. This finding highlights its suitability for lightweight neural architectures.

4. CONCLUSION

This paper proposes a lightweight super-resolution preprocessing model for JPEG compression tasks. On edge devices that cannot obtain high-resolution images, preprocessing enables the generation of high-quality images without requiring any modifications to the decoding end. We introduce a JPEG compression-aware training strategy, utilizing a differentiable JPEG codec for gradient backpropagation during the training strategy, and employ structural reparameterization and weight-sharing strategies to increase model depth and enhance multi-branch performance while reducing inference time. Comprehensive ablation studies validate the effectiveness of our modules and training strategies.

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